

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1}$ (1) $V_2 = \frac{1.01 \times 10^5 \times 125 \times 325}{1.01 \times 10^5 \times 297}$ $V_2 = 137 \text{ cm}^3$ (1)	1	1		2	2	
		(ii)		$n = \frac{PV}{RT}$ (1) $V = 1.6 \times 10^{-4} \text{ m}^3$, $T = 293 \text{ K}$ (1) $n = \frac{1.01 \times 10^5 \times 1.6 \times 10^{-4}}{8.31 \times 293}$ $n = 6.64 \times 10^{-3} \text{ mol}$ (1)	1	2		3	3	
	(d)			the student is incorrect because: H_2O contains 2 bonding pairs and 2 lone pairs of electrons (1) BeH_2 contains 2 bonding pairs only (1) therefore different shapes since electron pairs arrange themselves to be as far apart as possible/ different number of electron pairs (1)			3	3		